SCHOOL OF ARCHITECTURE, COMPUTING & ENGINEERING

Computing & Engineering CN7039 Software Engineering

CW REPORT

Group Z

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DELIVERABLE D1: COMPANY BACKGROUND

1.BRIEF OVERVIEW

Organization: Amazon fulfillment center DRM2(Grays, Essex, UK)

Sector: E-commerce logistics and Warehouse management

Market/Customers: Business-to-Consumer (B2C) serving customers with fastest delivery promises.

Location: Warehouse in Grays, working everyday with many associates across multiple staff.

Strategy: Providing the best facility to customers which is focused on speed minimizing the redundancy of deliveries. The warehouse is based on the primitive application based on older techniques for the cart system and delivery. Our goal is to provide the best service for faster and safer deliveries to customers.

2. THREE PROCESSES NOT WELL SUPPORTED BY CURRENT IS/IT

1.CART ASSIGNMENT AND WORKFLOW MANAGEMENT-

To get a wheeled cart by an employee so he can begin to collect items for deliveries. The carts management is poorly done because employees sometimes find it hard to get the carts.

Problem: There is no proper management system. Employees wait in line for the carts to become available. This is leading to wasting time on deliveries**.**

Why? Waiting time is a pure waste for employees and deliveries, which slows down the whole process. The company generally supervises the time spent by the employees and the number of carts he has done in the whole shift. Consequently, it is useful for the employees to save time finding carts.

2. REAL-TIME CART TRACKING SYSTEM

The location of carts in the warehouse is unknown. Carts can be anywhere in the warehouse and might be just dropped by the drivers for the next delivery

Problem: Managers have no info on live maps to check cart locations. They can see the carts' current situation. Also. The supervisors have no access to such information.

Why? Delivery company has no GPS on its carts. The administration has no access to check the location, and it leads to poor decisions. When the drivers need the cart, they also struggle to find them

3. CART TURNAROUND AND COMMUNICATION

When a Cart is full and needs to be unloaded, Cart is idle and Cart is not available so employees can use it.

Problem: Employees unload the carts themselves, but the system does not efficiently work when a cart needs service. Managers and associates communicate through multiply channels (walkie- talkie, shouts).

Why? When employees do jobs outside their role (like unloading), it interrupts the workflow. The application has required options for it to update in the system.

DELIVERABLE D2

1. **SELECT AND JUSTIFY THE FOCAL PROCESS/SYSTEM (AGREED WITH TUTOR).**

   a. The pickup cart process should be encompassed with live tracking for the warehouses associates and the drivers. Even though the associates have the timing to complete the carts in 7 min, the drivers have no tracking system for the carts. Overall, the time wasted by the employees results in company losses.

   b. Due to this the operation cost of the amazon gets huge loss because the associates get the carts after 15-20 minutes. During peak seasons like prime day and Christmas eve.

   c. The warehouse associates were never able to achieve the required targets for carts, i.e. 15-17 per associate as suggested by the supervisor manager. This problem results in bad performance while employees are not directly responsible for it.

2. **REQUIREMENTS COLLECTION TECHNIQUES**

   Technique 1: Observation

   a. During the peak working hours of warehouse at amazon the associates get the same cluster during their pickups due to which there get a traffic jam in cluster, and it get hard to move fast within the cluster for the pickups so the manager should do the traffic load balancing so that they should not get the bottleneck jam in the cluster.

   Technique 2: Interviews

   a. Warehouse associates: Understanding their pain points like manual unloading carts and storing the items in assigned packages.

   b. Supervisor: the supervisor must allocate the resource, and he should not have the lack of visibility of carts so that Associates achieve their targets smoothly.

   c. Manager: To grasp the strategic impact, budget constraints, and overall process goals.

3. STAKEHOLDERS AND ROLES IN DEVELOPMENT AND USE

a. Warehouse Associates:
a.1 Role in development: providing feedback on cart unavailability and manual unloading.
a.2 Roles in use: System supports receiving cart assigned and reports on issues (e.g., carts are full/unavailable).
b. Supervisor:
b.1 Role in development: they should be able to create a report and manage the workflows with proper approval.
b.2 Role in use: they should use a system dashboard to manage the associates so that they can check the work is running smoothly and manage exceptions.

c. Operation manager:
c.1 Role in development: they should provide a high level of goals and the required budget.
c.2 Role in use: they should be able to analyze high-level reports for overall efficiency and ROI.

d. IT support team:
d.1 Role in development: They should be able to develop a dashboard, i.e. free from bugs.
d.2 Role in use: maintain the hardware and software, e.g., sensors, scanners, servers, and databases.
e. System analyst
e.1 Role in development: we should design the plan and implement the plans. To solve the problems of carts, delay
The role is to prepare blueprints for the carts issue.

4. **DETAILED TASKS, INPUTS/OUTPUTS, REPORTS, SOFTWARE/ HARDWARE**

   a. **Detailed task:** Associate logins
      Inputs: Associate ID/Badge
      Outputs: System Authentication: Associate is "in queue" for the cart
      Reports: Easily accessible
      Software/Hardware: Login hub palm device scanner

   b. **Detailed task:** System assigns cart
      Inputs: Associate location, Available cart locations
      Outputs: Nearest associate assigned for the cart by notification
      Reports: Realtime alert
      Software/Hardware: Associate handheld device

   c. **Detailed task:** Associate logins
      Inputs: Associate ID/Badge
      Outputs: System Authentication: Associate is "in queue" for the cart
      Reports: Easily accessible
      Software/Hardware: Login hub palm device scanner

   d. **Detailed task:** Associate marks Cart Full
      Inputs: Cart full signal
      Outputs: Status changes to full for unloading. Location details sent for unloading
      Reports: Real time alert to unload cart
      Software/Hardware: Handheld device

   e. **Detailed task:** Unloader empties cart
      Inputs: Unloader ID, Cart ID scan at unload station.
      Outputs: Cart status changes to empty
      Reports: Picking hours of per associate for productivity reports
      Software/Hardware: Barcode readers for unloading

f. **Detailed task:** System Monitoring
Inputs: Continuous Status and location of all carts.
Outputs: Manager dashboard real-time assessment of carts and associates.
Reports: Operational efficiency report average cart turnaround time, idle time metrics.
Software/Hardware needs: CMS Dashboard Software.

5. MODELS: CURRENT-STATE USE CASE DIAGRAM AND ERD.

USE CASE DIAGRAM OF THE SYSTEM

This use case diagram depicts the current state of the warehouse management system. Where the entities and their use cases are defined as shown in the diagram.

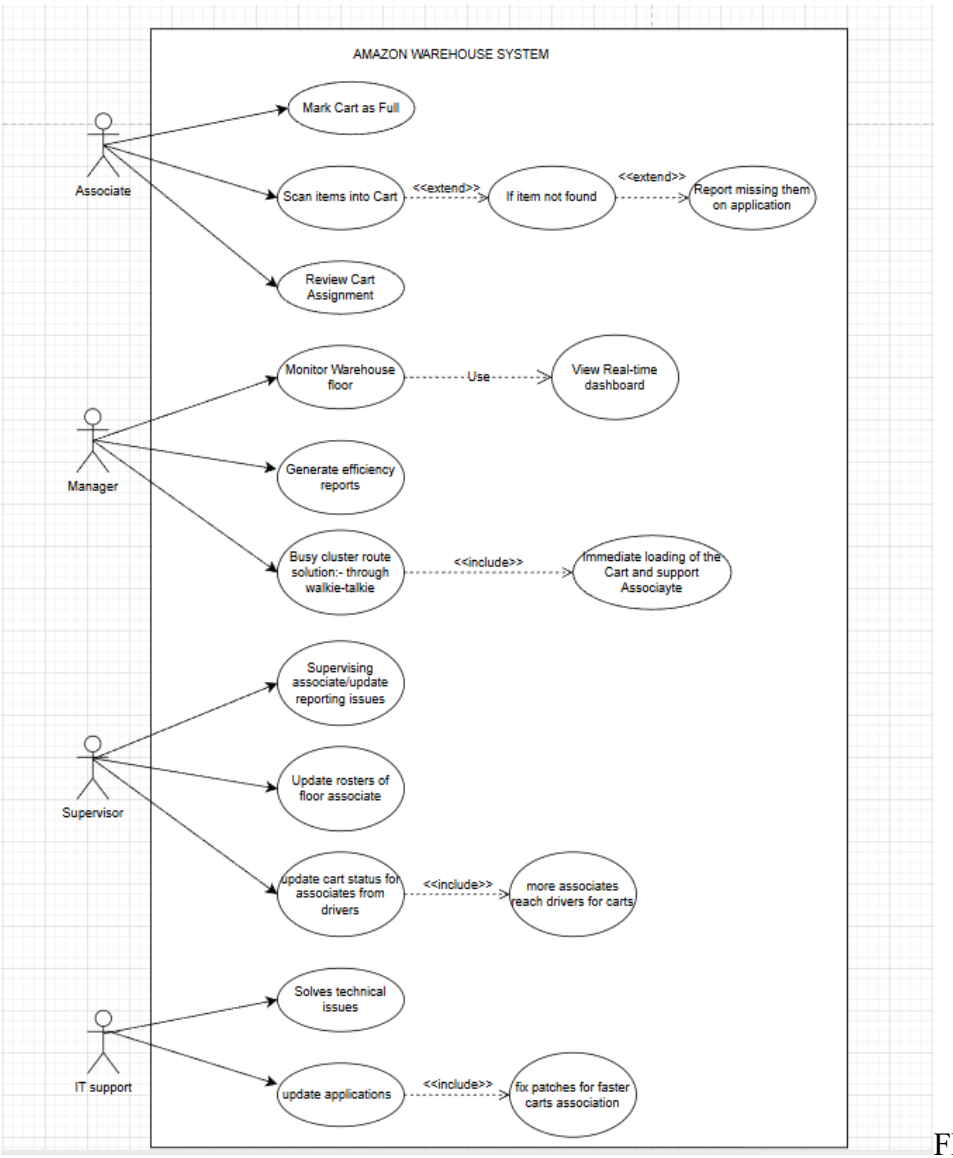

FIGURE 1

ERD diagram of the system

The ERD diagram of the warehouse management system in the current situation where the entities relations are shown in figure 2. where primary key and foreign key are assigned for the relations.

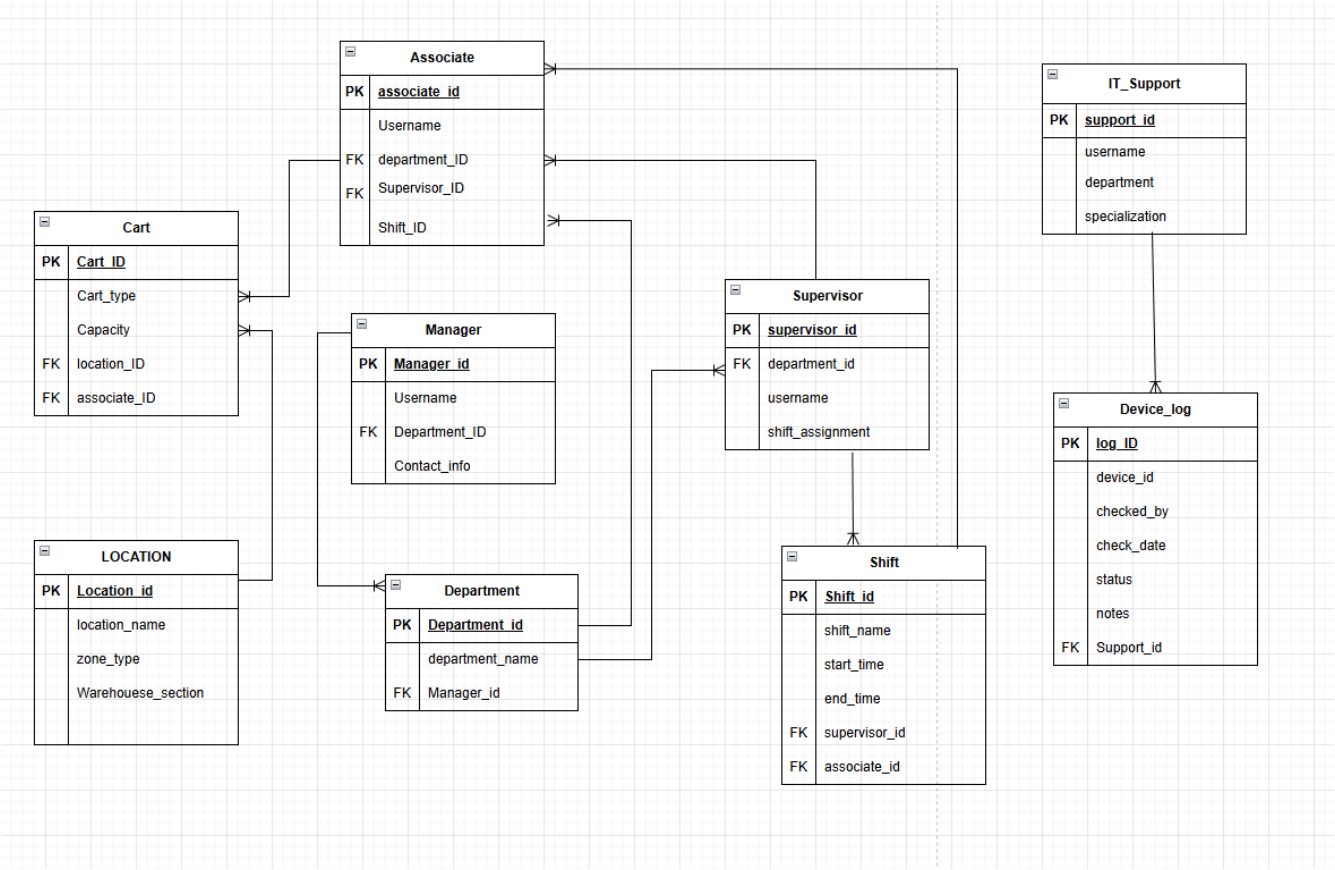

FIGURE 2

TERMINOLOGY AND EXPLANATIONS:

Associate- Warehouse workers who uses carts UserID- Associate ID
Username – person's name

Department
Manager- Overall warehouse manager
ManagerID
Username- name of manager

Department
Supervisor- Shift supervisor
Supervisor- Supervisor ID
Router Details- Route details (work assignments) IT
Support- Technology support staff ITsupport-IT
Staff id

Department
Username Staging
Location
Staging ID-Location ID
Cart ID- which cart is in the location
Baggage- Current load
Cart- the wheeled carts used for moving packages
CartID - cart identification number
Baggage

6. PROJECT RISK ASSESSMENT (TABULAR)

Risk category	Description	Probability (1-5)	Impact (1-5)	Risk Score	Owner	Mitigation Strategy
Technical Integration failure	**Cart Management System fails to integrate with existing warehouse management system**	4	5	20	CRITICAL	**Early Engagement with Amazon IT. Choosing and API**
User Resistance	**Employees resist the new process, inclined toward old ways**	5	4	20	CRITICAL	**Involve users in the design phase.**
Data accuracy	**sensors or barcode scans provide**	3	4	12	HIGH	**Robust testing of hardware during**

	inaccurate data					**pilot. Implement a pilot program.**
Budget Overrun	**The project expands to include other warehouse processes, losing focus**	2	4	8	MEDIUM	**Clearly defined project charter signed by the sponsor. Strict change control procedures.**
Hardware Reliability	**Bluetooth beacons fail; handheld devices break in warehouse environment.**	3	5	15	HIGH	**Select android hardware. Extra spare inventory maintained on-site. Fast processed pre-configured devices**
Training insufficiency	**Inadequate training leads to low adoption and errors.**	4	3	12	HIGH	**Role based training. Video tutorials, on floor training.**
Scope Creep	**Project expands to include inventory**	3	3	9	MEDIUM	**Signed project charter with clear**

	management					**boundaries. Change request process requiring sponsor approval Regular scope review meetings.**
Network downtime	**Warehouse Wi-Fi fails while real time tracking**	2	4	8	MEDIUM	**Redundant mesh network design. Manual mode availability,**

RISK SCORING KEY:

Probability Scale:

1 = Very Unlikely

2 = Unlikely

3 = Possible

4 = Likely

5 = Very Likely

IMPACT SCALE:

1 = MINIMAL (MINOR DELAY, £1K)
2 = LOW (SMALL SCHEDULE SLIP, £1-5K)
3 = MODERATE (MISS METRICS, £5-20K)
4 = HIGH (MAJOR DEALYS, REPUTATIONAL DMAGE, £20-100K)
5 = CRITICAL (PROJECT FAILURE, SAFETY ISSUES,>£100K)

PRIORITY THRESHOLDS:

15+ = CRITICAL (IMMEDIATE ACTION REQ.) COLOUR RED
10-14 = HIGH (SENIOR MANAGEMENT ATTENTION) COLOUR YELLOW
5-9 = MEDIUM (MONITOR AND MANAGE) COLOUR GREEN
<5 = LOW (ACCEPT AND MONITOR) COLOUR BLUE

TOP 3 RISKS REQUIREING IMMEDIATE ATTENTION:

1. **TECHINCAL INTEGRATION FAILURE (SCORE 20)**
   Failure here means the system cannot work at all.
   Action required: Technical lead to complete troubleshooting.

2. **User Resistance (Score 20)**
   Technology fails if users are not using it appropriately
   Action required: Change manager to identify associates.

3. **Hardware reliability (Score 15)**
   Warehouse system failure would halt working if hardware is not available.
   Action required: IT support led to select suitable hardware.

DELIVERABLES D3

PROPOSED SOLUTION USE CASE DIAGRAM.

In the previous management system, the problems are rectified to make the system efficient. The required solution to reduce the cart issue, we have added the updating of cart status, and cart tracking which accessible for the **associate** and real time dashboard which is connected to **Manager** as well as **supervisor**. By this way it becomes easier for the associate to track the cart and check the location which results in faster deliveries and wasting less time in finding carts.

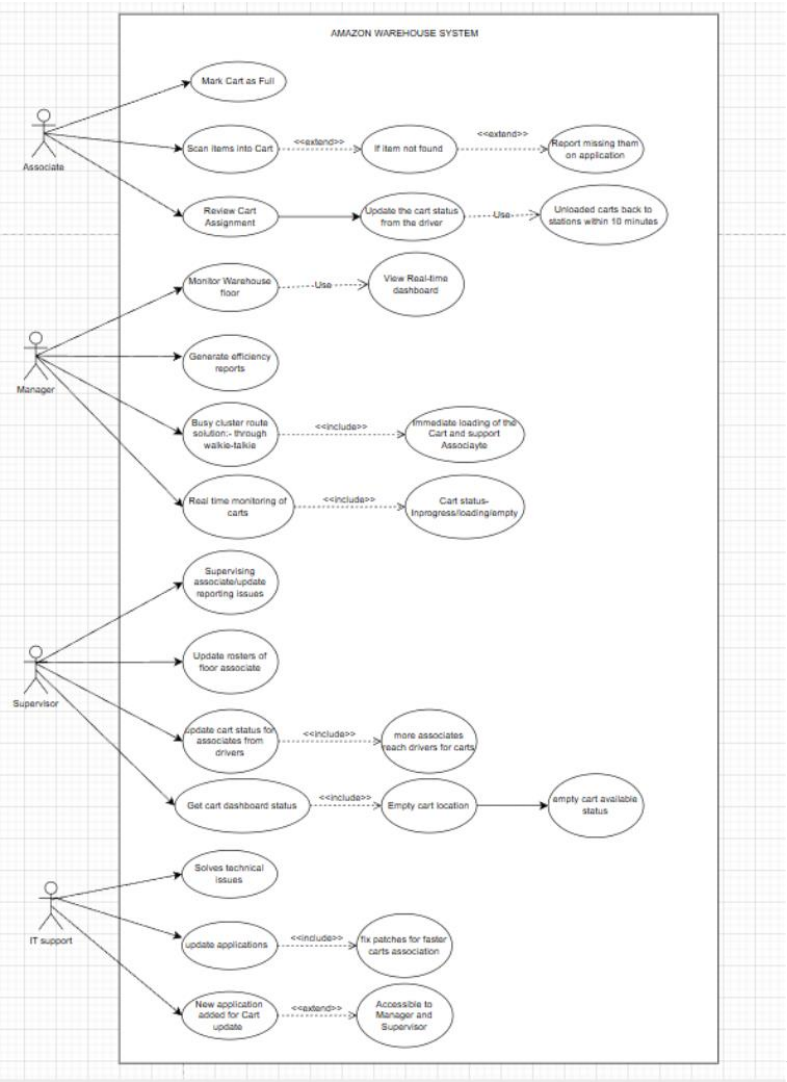

FIGURE 3

DATA FLOW DIAGRAM (CONTEXT + LEVEL(S) AS APPROPRIATE).

CONTEXT DIAGRAM LEVEL 0

- **Warehouse Associate:** Provides login, scan data, and cart status; receives assignments.
- **Warehouse Manager:** Provides query/report requests; receives reports and dashboard data.
- **Existing WMS (Warehouse Management System):** Provides task/picklist data; receives completion confirmations.
- **Supervisor -** update the roster and report issue, and get cart status from management system
- **IT support –** responsible for the system updates

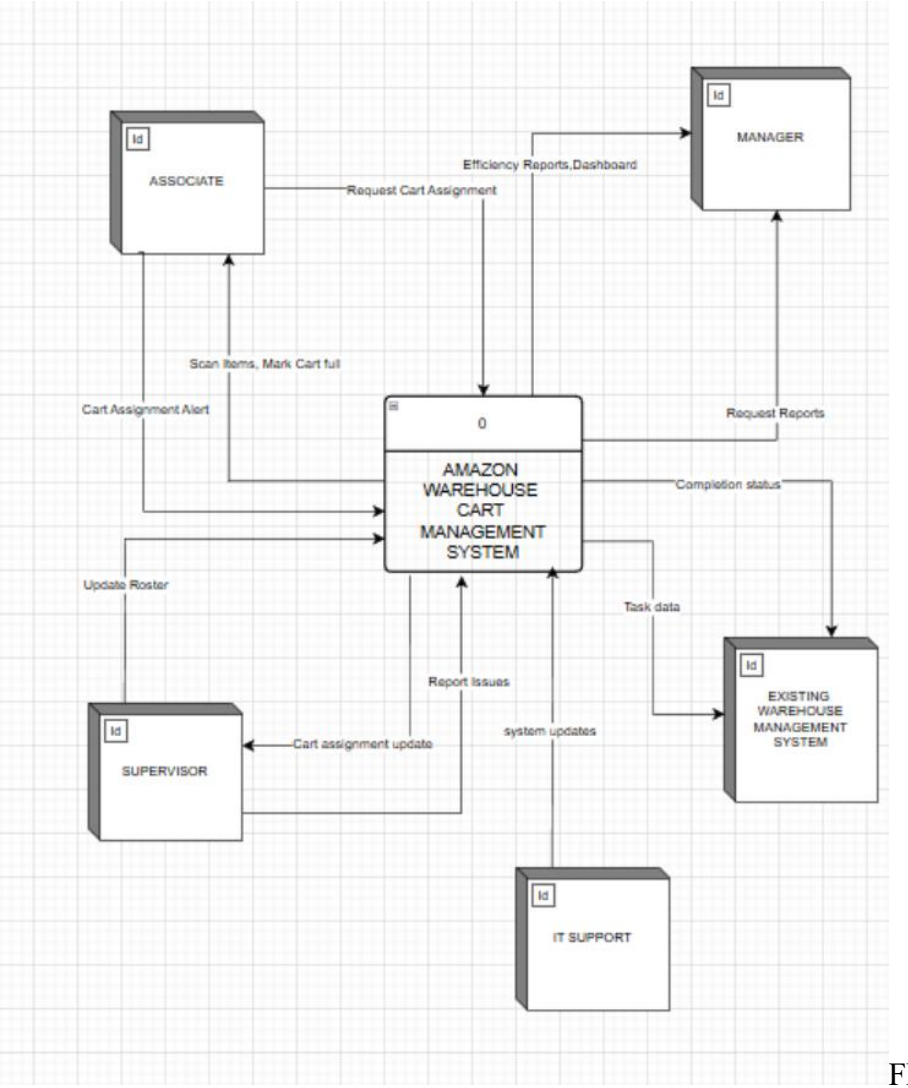

FIGURE 4

LEVEL 1 DIAGRAM

This breaks down the major processes inside the Cart Management System.

Entity:

Associate, Supervisor, Manager, IT support and Warehouse cart management system.

Processes:

1. 1.1 Manage Cart Assignment
2. 1.2 Track Picking Activity
3. 1.3 Monitor and Report
4. 1.4 System Maintenance
5. 1.5 Performance Analytics
6. 1.6 Checking employee data

Data Stores:

- D1 Employees: Stores Associate/Manager data.
- D2 Carts: Stores cart details (ID, type, status, location).
- D3 Assignments: Stores the active relationship between Associates and Carts.
- D4 Location: Stores the picklist from the WMS.
- D5 Performance log
- D6 Systems log
- D7 Tasks

The data flow diagram demonstrates the flow of the warehouse management system where Associate can easily access their data meanwhile, it has two processes managing cart assignment and track picking activity which have their own databases as well which are interconnected to both the processes to make it easy for the warehouse cart management system to update tasks and read/write in the database D7 tasks. Supervisors can monitor and report and make changes to the database. Although Manager check the performance analytics of the system. Conclusively The system maintenance is done by the IT support and in this way whole system is working

perfectly fine and efficiently. The major issues were solved and results with better performance and productive work environment.

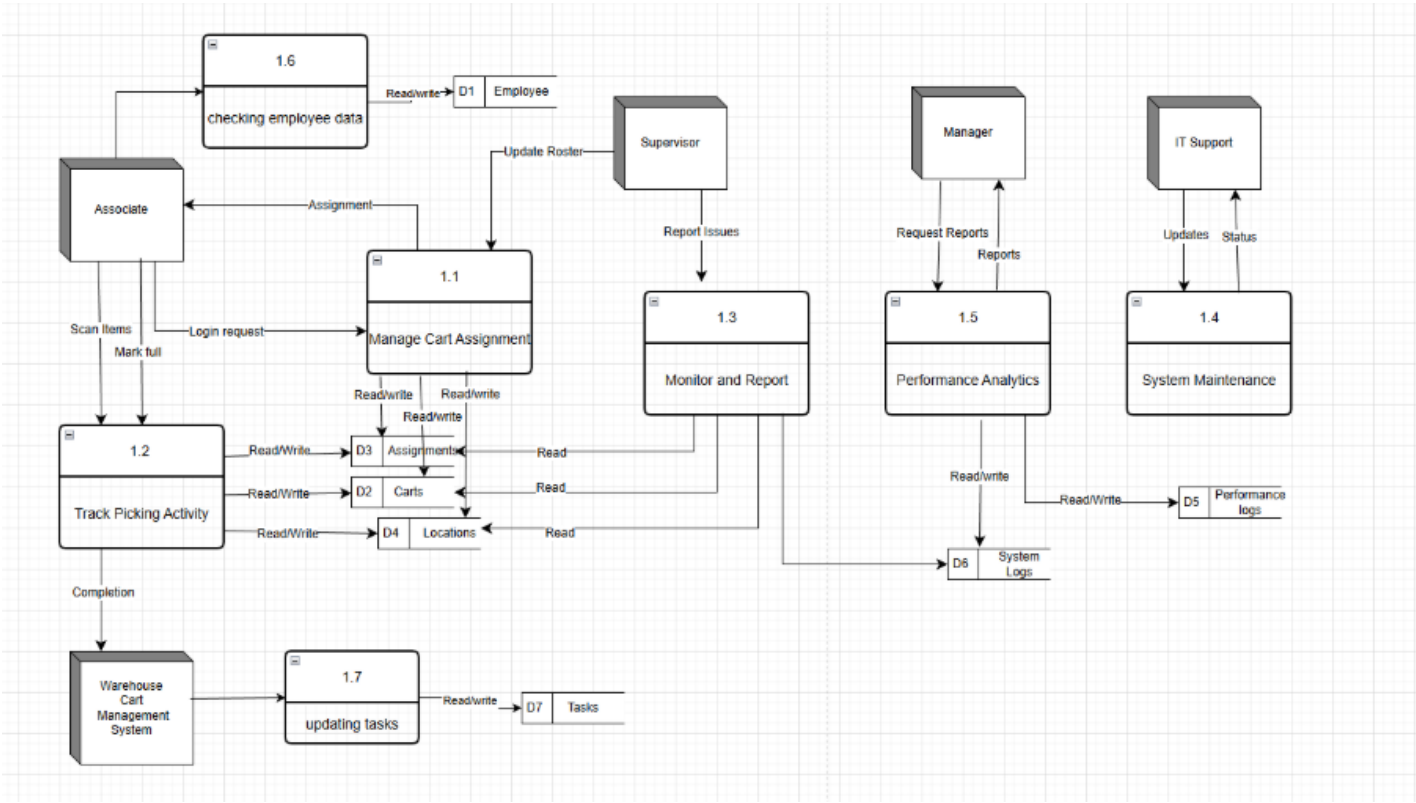

FIGURE 5

DATA STORAGE DESIGN USING A CLASS DIAGRAM.

The class diagram is the blueprint that demonstrate how the data is to be stored, accessed, structured and managed in the whole system. Here the Employee entity has its own attributes and behaviour which is one to many relationships with assignment, performance log, and system log. Whether the Shift and the Department make one to many relationships with employee class. The Task entity has one to one relation with assignment. Also, the Cart has one to many relations with assignment. Location has one to many relations with the cart. The entities have defined their attributes and the behaviours in the diagram.

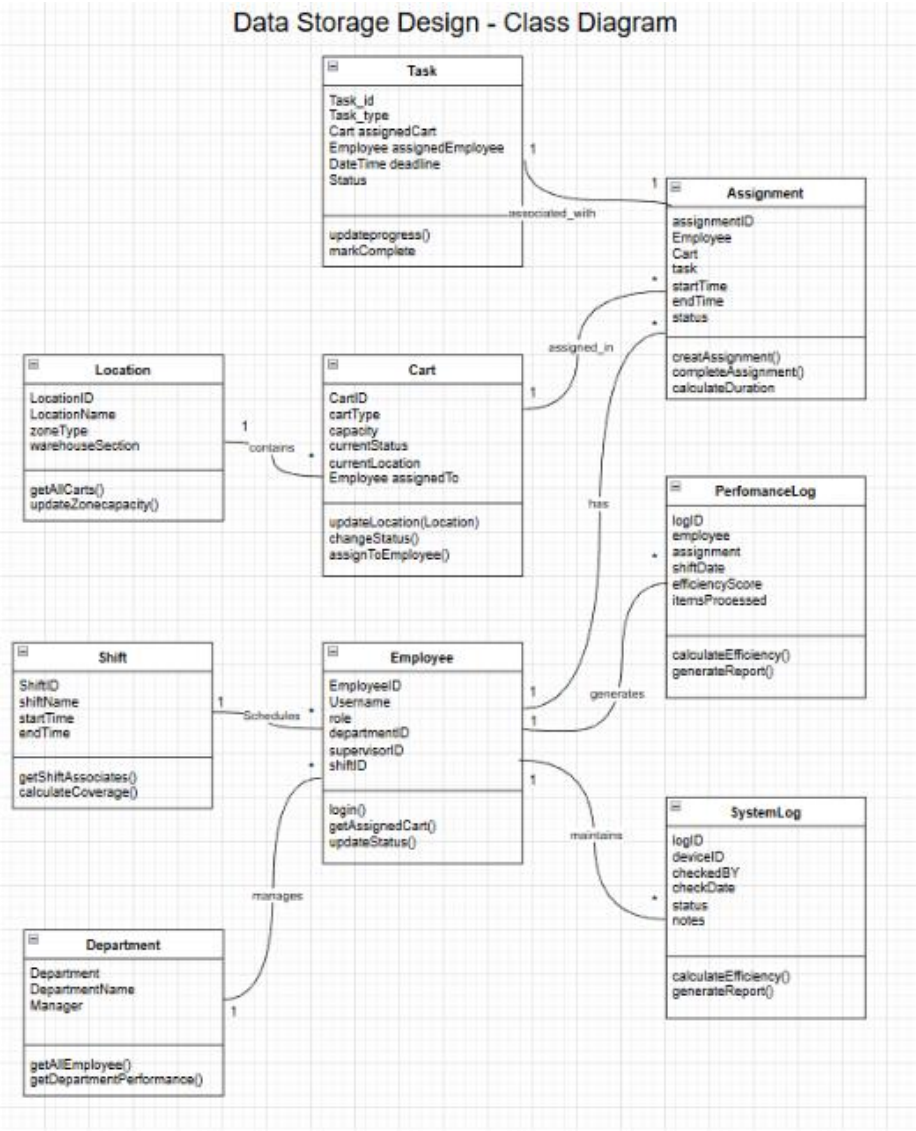

FIGURE 6

SYSTEM CONSTRUCTION (IMPLEMENTATION) ARCHITECTURE.

We have chosen the Internet- and Intranet-based Architecture for the implementation of the warehouse management system. In which presentation layer is managed by the client pc where the Manager, Associate interface on handheld device, Supervisor console is on the tablet and IT support portal is on the admin pc. The authentication is in the web server for the login of the users. The application server is there for managing the application logics. The Data layer has the database server, and the data manipulation logics can be only executed in this layer which is safe and secure from the attackers. This choice is best for the security purpose of the system as the unwanted cyberattacks and bugs try to get in the system, and they cannot get into to the system because of the multilayer distribution of the servers where every layer has its own dedicated server.

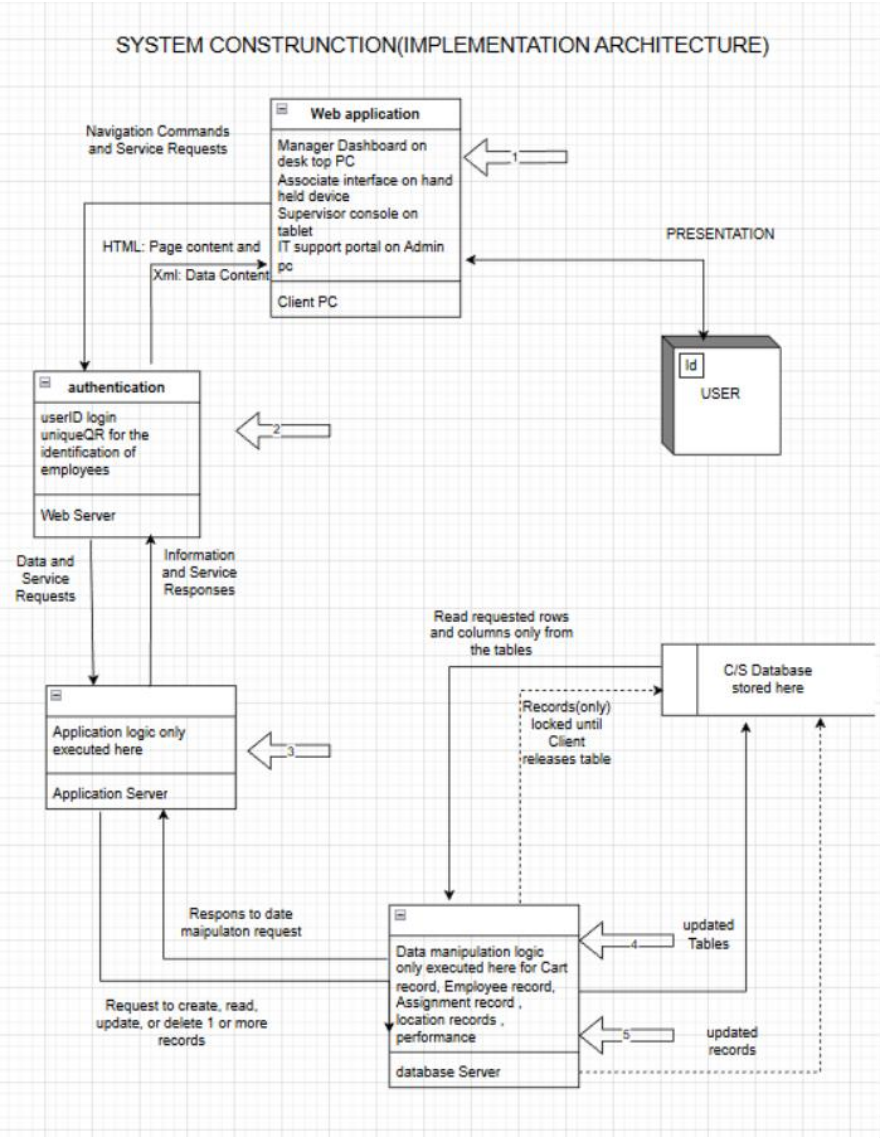

FIGURE 7

Evaluation

Strengths of the Proposed System

The Cart Management System states the major inefficiencies identified at the Amazon Warehouse at Grays London UK. The proposed solution is well-structured and demonstrates a crystal-clear understanding of the problem domain.

1. Target Problem Resolution: - The system specifically tackles the three critical issues: cart assignment inefficiency, lack of real-time tracking, and poor communication for cart turnaround. The introduction of the automated cart assignment, GPS based tracking and digital "Cart Full" notification system are a logical and effective counter measures approach.

2. Comprehensive Analysis: - The deliverables show thorough analysis, moving effectively for the problem identification in phases (D1) states the through requirements gathering, (D2) states modeling, (D3) Proposed to technical design. The use of UML diagrams, Use Case, ERD, Class Diagram, and Data flow diagram provides a clear, standardized visual representation of both current and the proposed state system.

3. Practical Architecture: - The proposed 3 tiers (Presentation, Application and Data) internet/intranet architecture is justified for the warehouse environment which allows for robust central management while supporting various client devices like handheld, tablets and PC ensuring the accessibility to all the stakeholders.

4. Risk-Aware Planning: It proactively identifies critical risks of life technical integration failure, user resistance, assigning ownership and mitigation. This helps a mature approach to project management.

5. Data-Driven Design: The class diagram and DFD level 1 logically structure the system talks about the data entities (Cart, Assignment, Employee) and processes talk about the (Manage Cart Assignment, Track Picking Activity) which ensure that all the information flows are captured and managed.

Areas for Development and Consideration

1. Implementation Specificity: D3 shows the high-level design. This is going to help in further specifications, such as API protocols for integration with the existing Amazon Warehouse Management system for data synchronization and specific hardware models like scanners.

2.Scalability and Performance: The architecture diagram shows the Database and Application server which is applied for the large-scale operations like an Amazon FC which considered for the load balancing, database replication and cloud-based scalability

Which is going to be helpful in the peak season like Amazon Prime Day, Boxing Day, and Christmas Seasons.

4.Security Considerations: - The security could be expanded based on authorization levels (roles-based access control – RBAC), data encryption for transmission and audit trails for the security compliance.

5.Prototyping & Pilot Validation: The success is based on the user's adoption. It would be developing a functional prototype or running controlled pilot in one zone of the warehouse. This would help in validating the design with the real users (Associates, Supervisors, Managers) before full-scale deployment.

CONCLUSION

This report elaborated the analysis and design for a **new Cart Management System** at the Amazon Warehouse DRM2 Garys United Kingdom. It notifies operational bottlenecks in the current cart assignment, tracking and communication processes, which lead to waste time, reduced productivity, and increased operation costs.

The Suggested solution like introducing an automated, data-driven system featuring real-time cart tracking, intelligent assignment algorithms and streamlined digital workflows. This Project deals with solutions to eliminate the associates' wait times, provides management with actionable visibility, and optimize cart utilization. The structured roadmap shows the stakeholders' analysis and requirement collection through the architecture design – which demonstrates strong software engineering principles. The primitive problems showed with the use case and Erd diagram and the following solutions are efficiently explained with the help of several diagrams like use case, ERD, data flow diagram class diagram and system implementation architecture diagram. Conclusively, the whole coursework is the complete solution for the betterment of the care management of the warehouse system.

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